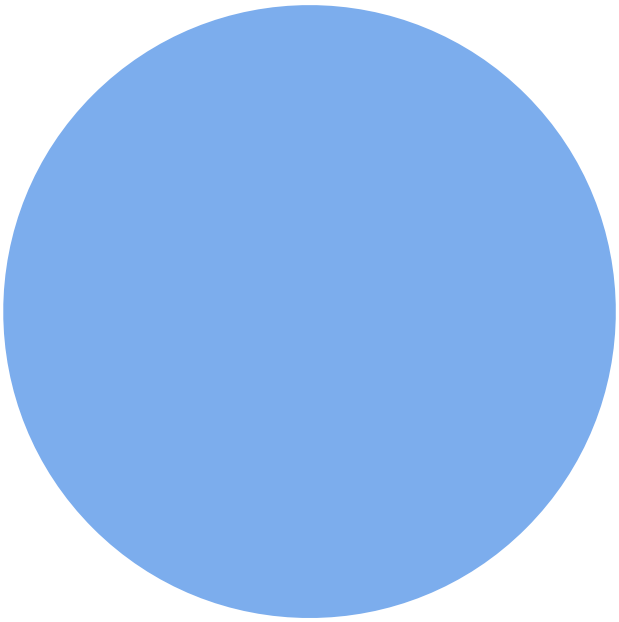
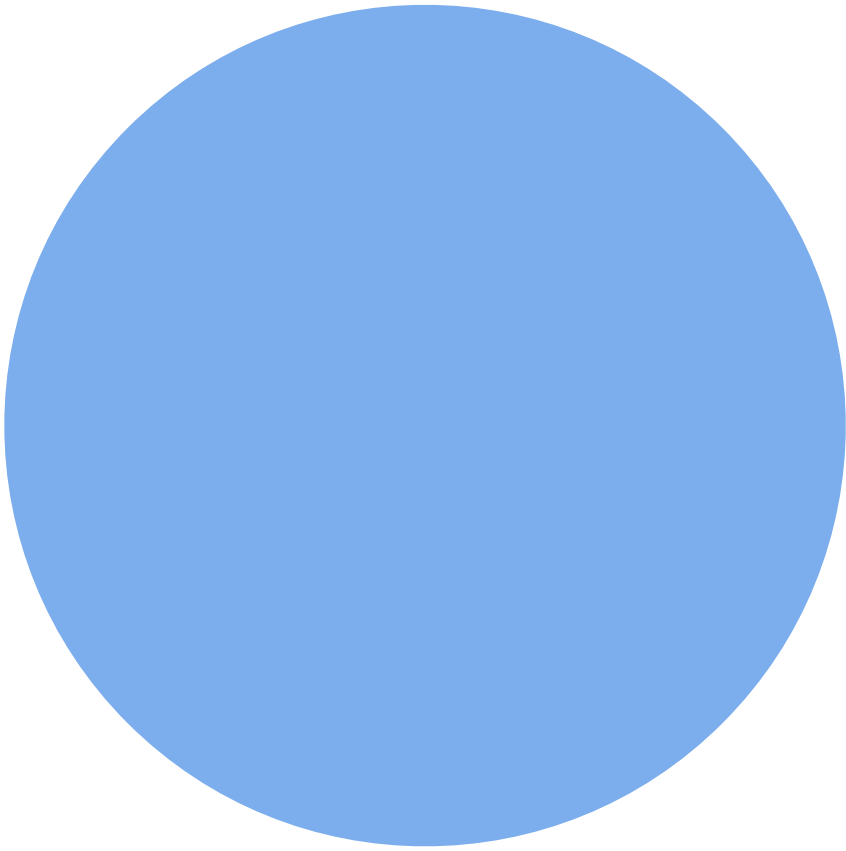
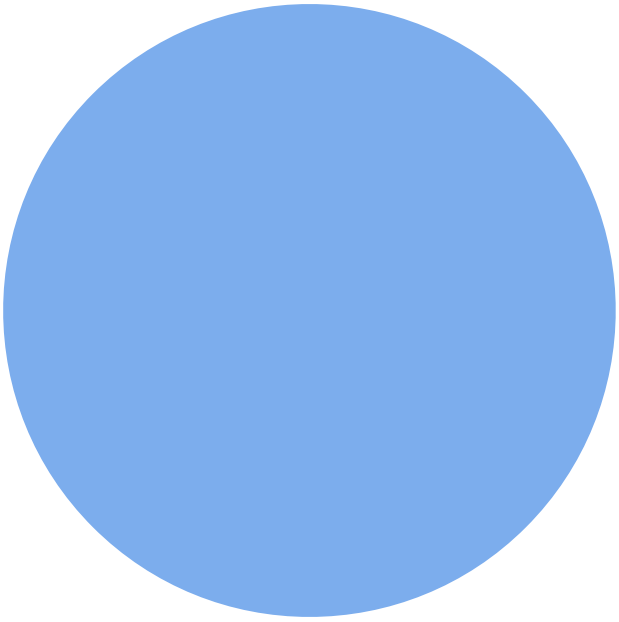
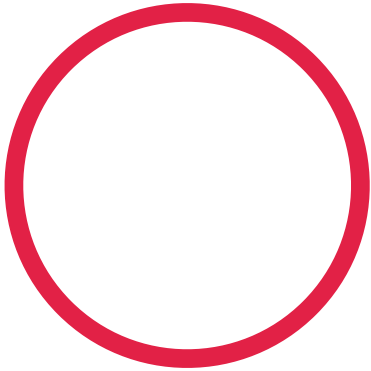


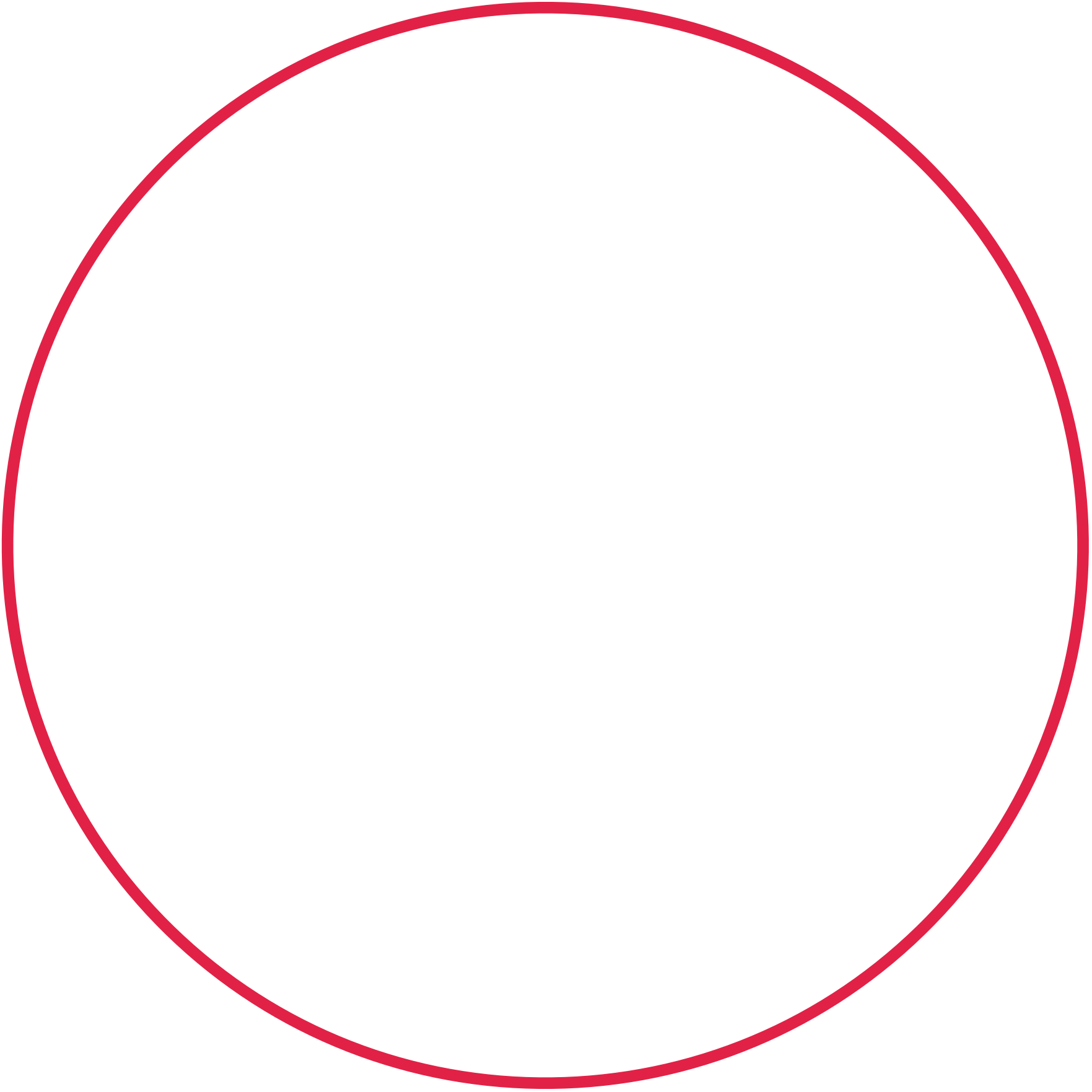
Q3: How many circles are simultaneously tangent to 3 given circles in the plane?

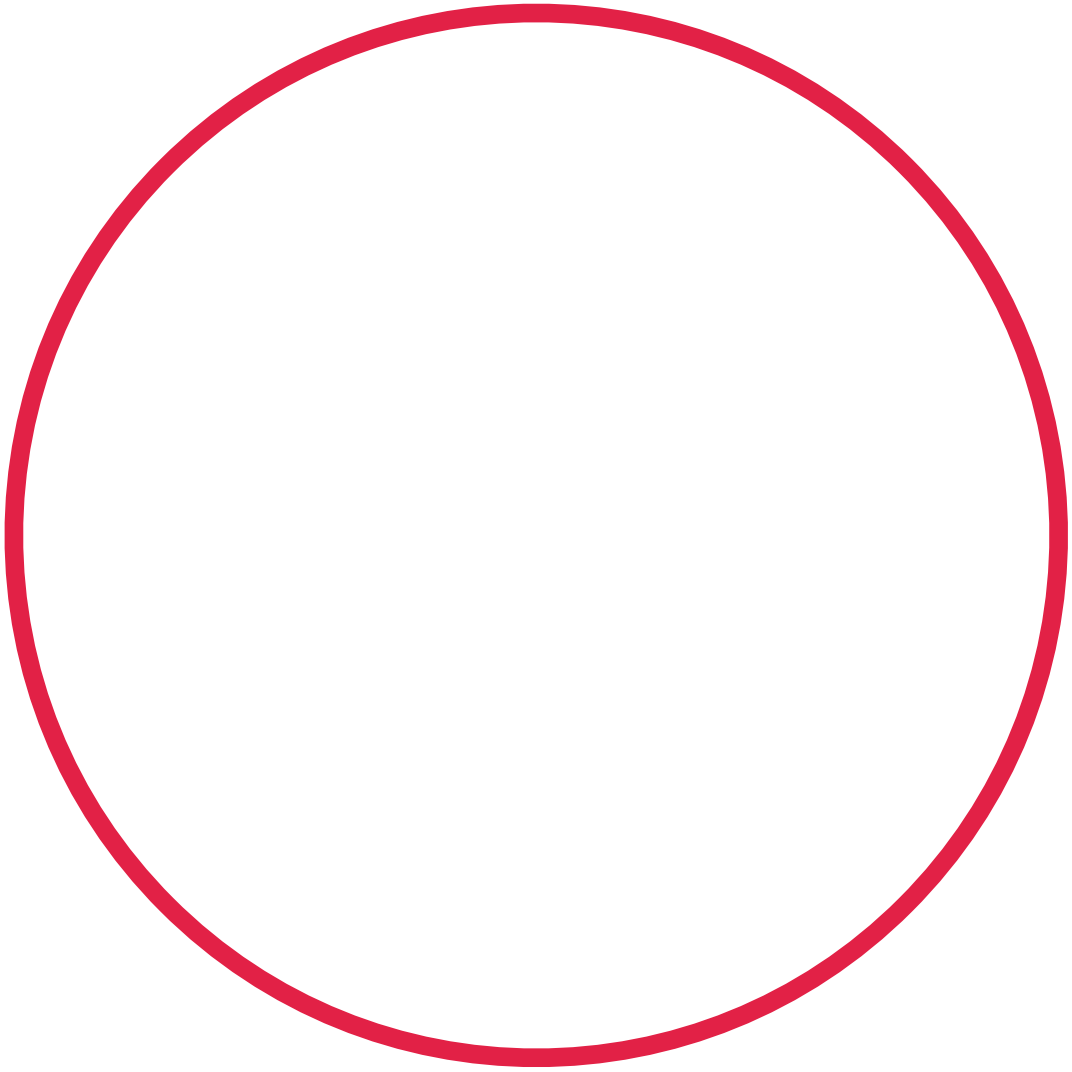


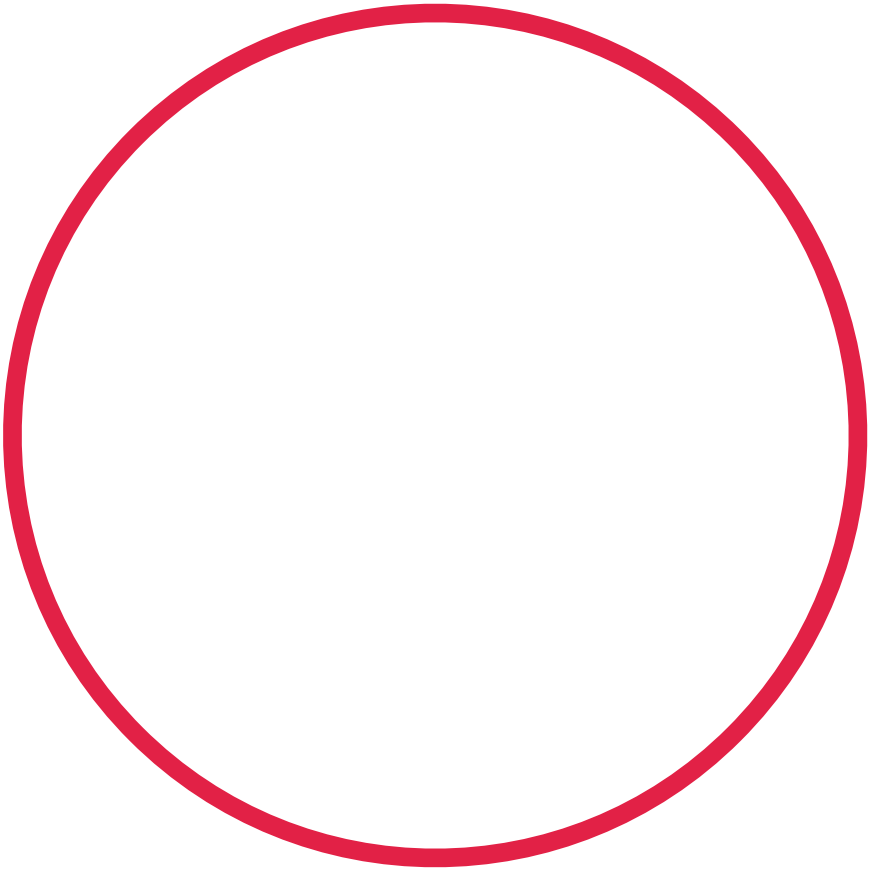


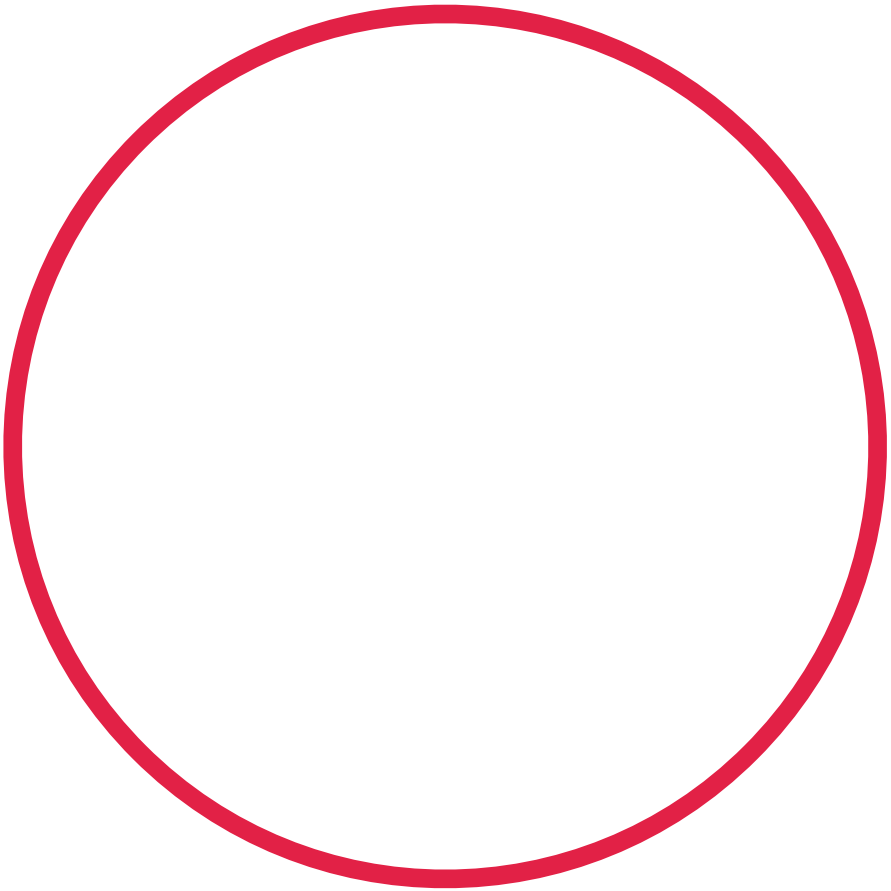


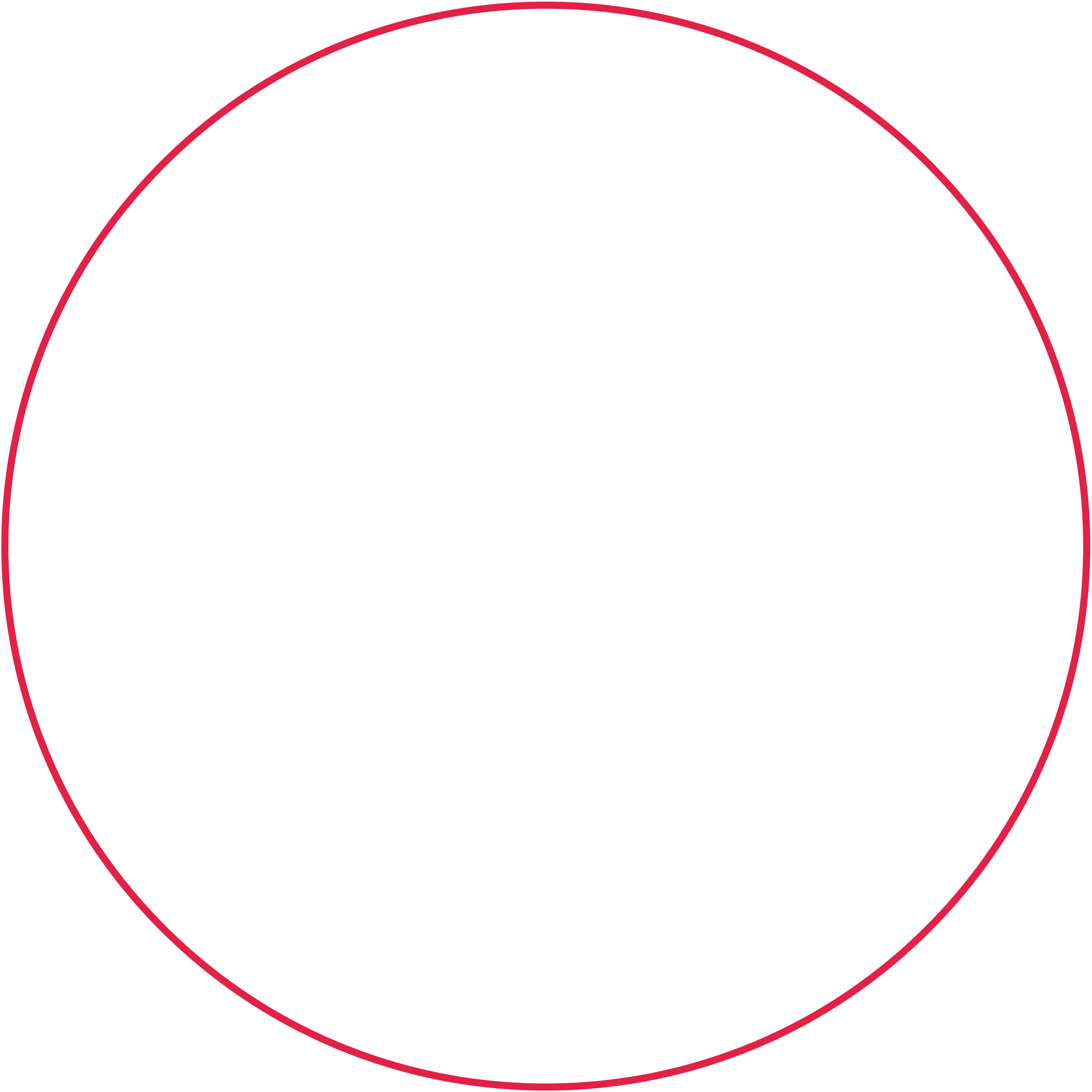


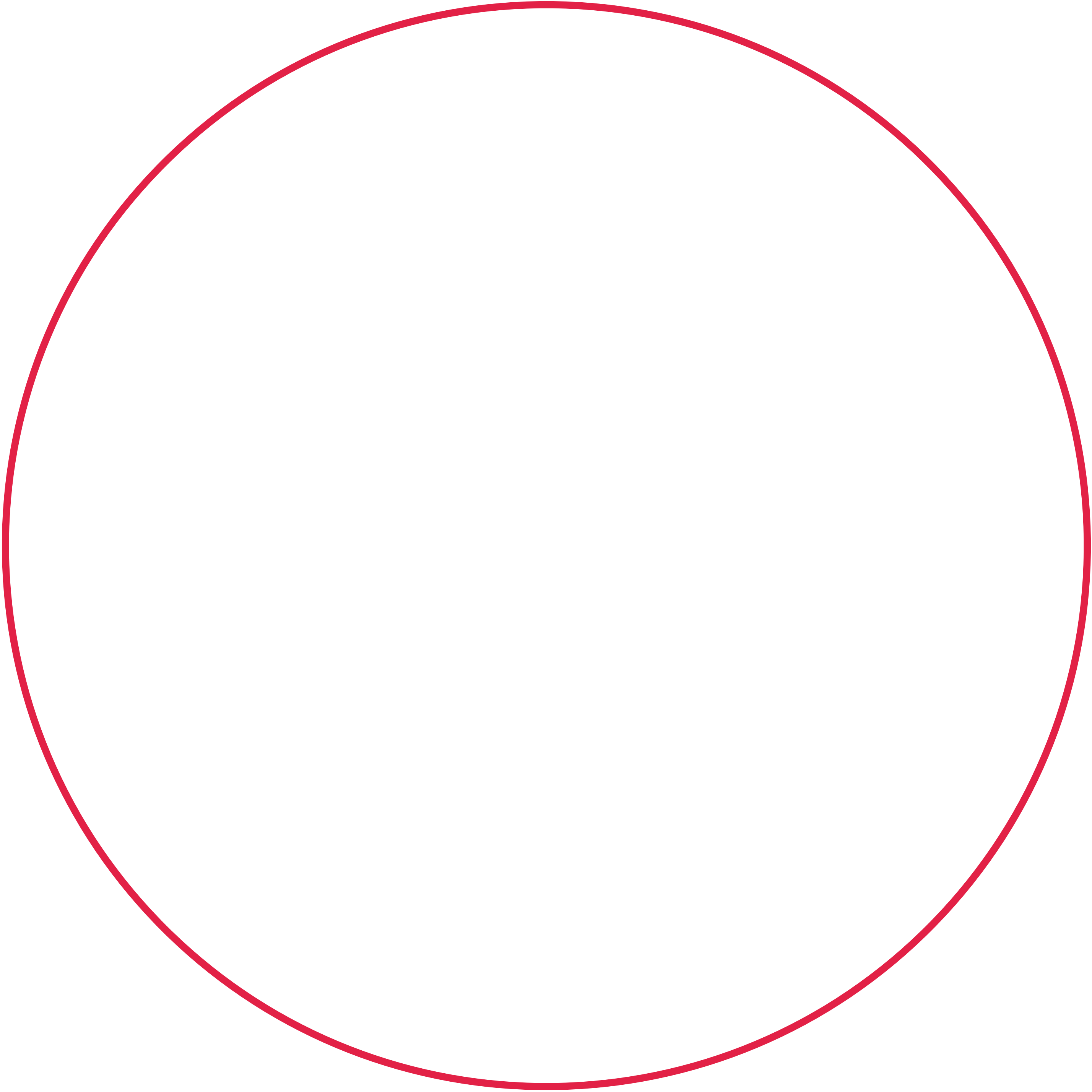


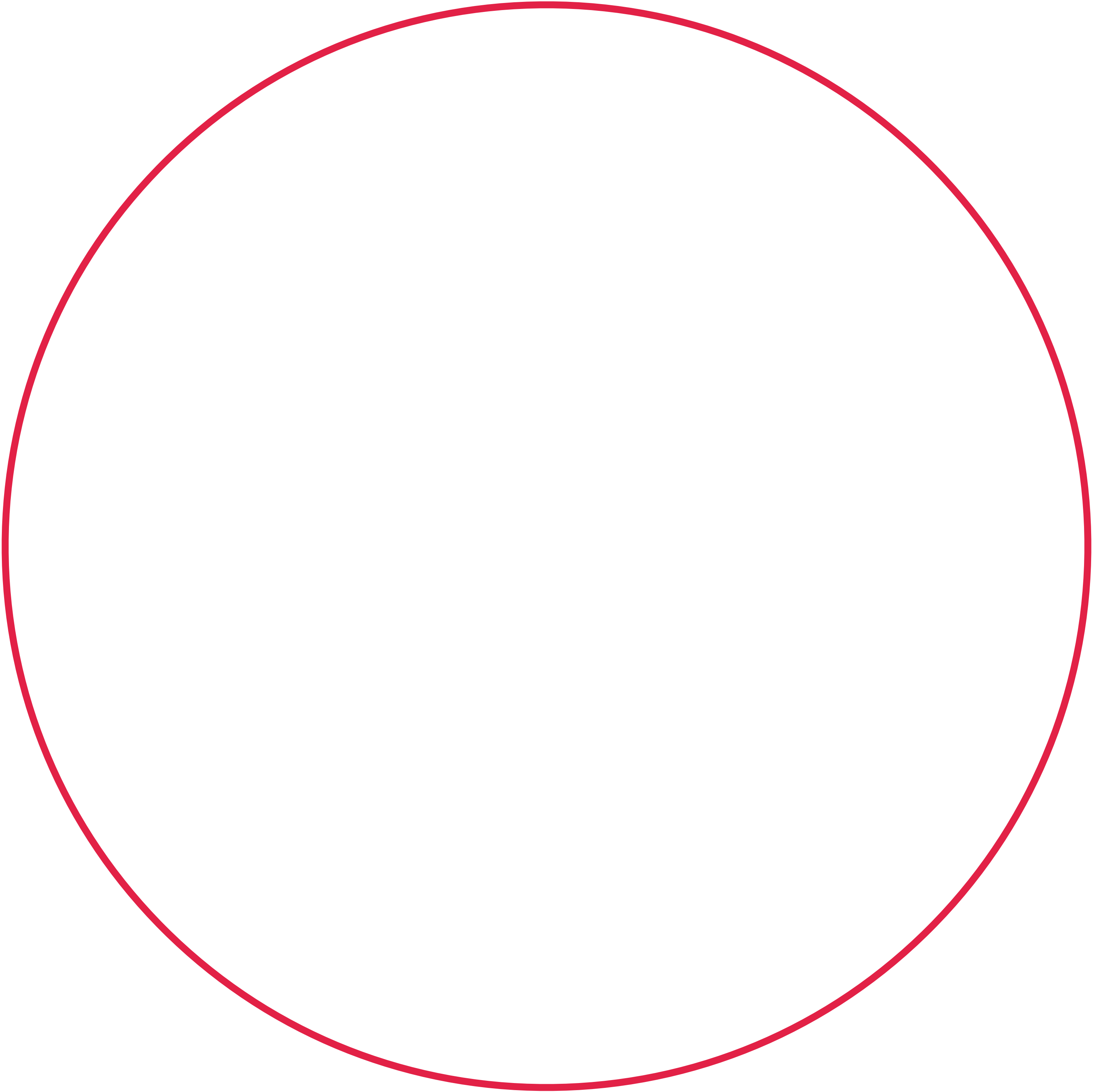












The Circles of Apollonius

0, 1, 2, 3, 4, 5, 6, 8 over $\mathbb{R}$

8 over $\mathbb{C}$

Q3: How many circles are simultaneously tangent to 3 given circles in the plane?

The Circles of Apollonius

8 over $\mathbb{C}$

0,1,2,3,4,5,6,8 over $\mathbb{R}$

Q4: How many lines on a smooth cubic surface?

$$\text{Clebsch cubic } x_1^3 + x_2^3 + x_3^3 + x_4^3 = (x_1 + x_2 + x_3 + x_4)^3.$$